

Parallel Processing

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Parallel Processing

Parallel processing is a type of computation where many calculations or the execution of processes are carried out simultaneously.

Large problems can often be divided into smaller ones, which can then be solved at the same time.

There are several different forms of parallel processing: pipelining, instruction level parallelism, parallel processor architecture etc.

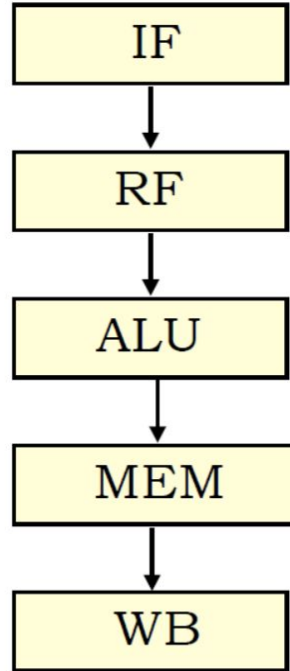
Pipelining

It is a process in which several storages of the CPU are used to execute more than one instruction concurrently.

We'll study the classic 5-stage pipeline:

- 1. Instruction Fetch (IF) stage:** Maintains PC, fetches instruction and passes it to RF.
- 2. Register File (RF) stage:** Reads source operands from register file, passes them to ALU.
- 3. ALU stage:** Performs indicated operation in ALU, passes result to MEM.
- 4. Memory (MEM) stage:** If it's a Load instruction, use ALU result as an address, pass mem data (or ALU result if not Load) to WB.
- 5. Write-Back (WB) stage:** writes result back into register file.

Pipelining



Normal Execution of Instructions

Instruc tions	Clock Cycle																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
I ₁	IF	RF	ALU	MEM	WB															
I ₂						IF	RF	ALU	MEM	WB										
I ₃											IF	RF	ALU	MEM	WB					
I ₄																IF	RF	ALU	MEM	WB

Normally for I₁ to I₄ instruction needs $4 \times 5 = 20$ clock cycle.

Pipelining

Instructions	Clock Cycle																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
I ₁	IF	RF	ALU	MEM	WB															
I ₂		IF	RF	ALU	MEM	WB														
I ₃			IF	RF	ALU	MEM	WB													
I ₄				IF	RF	ALU	MEM	WB												

But with the use of pipeline, it is done with 8 clock cycle. Note that these instructions don't have any dependency (Data/Control).

Pipeline Diagram

Pipeline diagram represents pipeline utilization over time.

Consider this instruction sequence:

```
LOAD R1, [4]
LOAD R2, [8]
SUB R6, R7
XOR R9, R10
MUL R12, R13
ADD R15, 1
```

Stages	Cycles									
	1	2	3	4	5	6	7	8	9	10
IF	LD	LD	SUB	XOR	MUL	ADD				
RF		LD	LD	SUB	XOR	MUL	ADD			
ALU			LD	LD	SUB	XOR	MUL	ADD		
MEM				LD	LD	SUB	XOR	MUL	ADD	
WB					LD	LD	SUB	XOR	MUL	ADD

Flynn's Taxonomy

A taxonomy first introduced by Flynn in 1972 is still the most common way of categorizing systems with parallel processing capability. Flynn picked two characteristics that he considered essential:

1. The number of instruction streams and
2. The number of data streams.

Flynn's Taxonomy

Flynn proposed the following categories of computer systems:

1. Single instruction, single data (SISD)
2. Single instruction, multiple data (SIMD)
3. Multiple instruction, single data (MISD)
4. Multiple instruction, multiple data (MIMD)

Flynn's Taxonomy

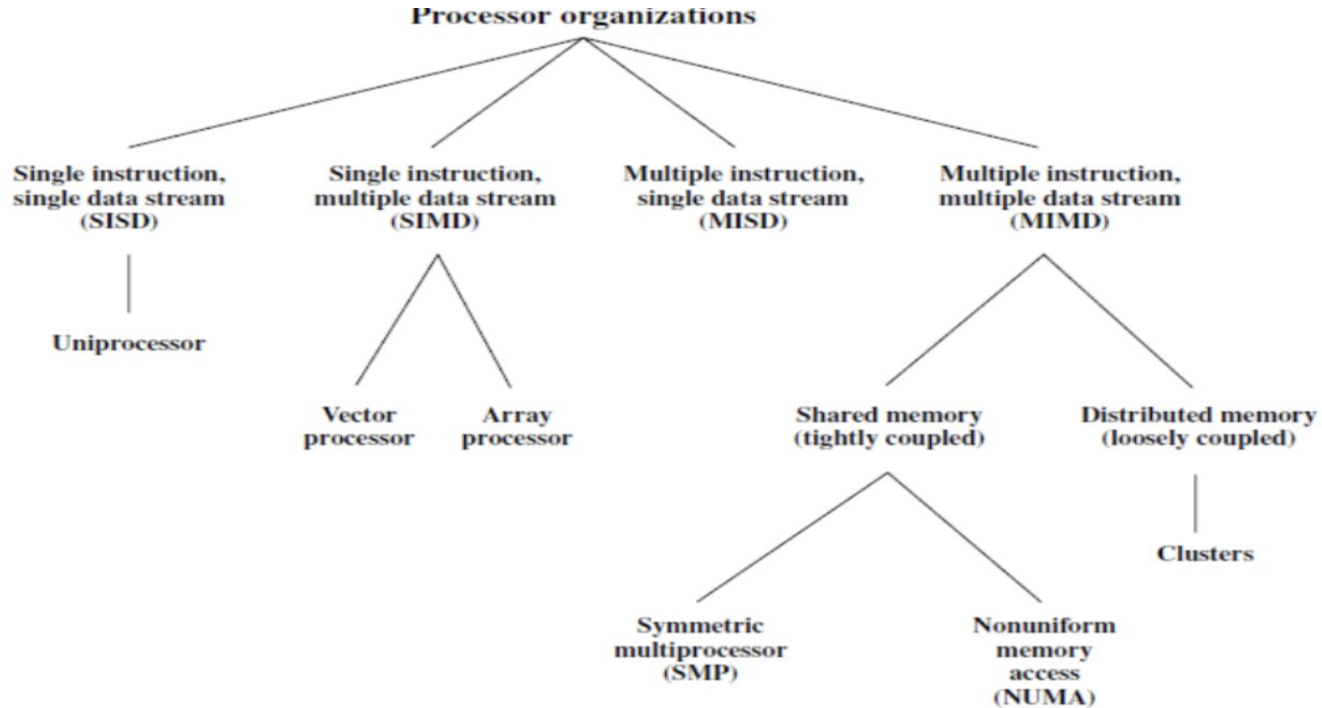


Figure: Taxonomy of Parallel Processor Architectures.

Flynn's Taxonomy

Flynn proposed the following categories of computer systems:

1. Single instruction, single data (SISD) stream: A single processor executes a single instruction stream to operate on data stored in a single memory.

For example, all traditional uniprocessor computers (having only one CPU) fall in this category from personal computers to large mainframes.

2. Single instruction, multiple data (SIMD) stream: A single machine instruction controls the simultaneous execution of a number of processing elements on a lockstep basis. Each processing element has an associated data memory, so that each instruction is executed on a different set of data by the different processors.

For example, vector and array processors fall into this category where processors with one instruction unit that fetches an instruction and then commands many data units to carry it out in parallel, each with its own data.

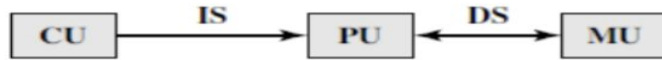
Flynn's Taxonomy

3. Multiple instruction, single data (MISD) stream: A sequence of data is transmitted to a set of processors, each of which executes a different instruction sequence. This structure is not commercially implemented.

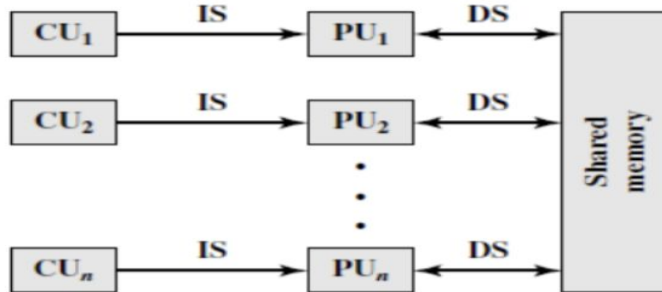
4. Multiple instruction, multiple data (MIMD) stream: A set of processors simultaneously execute different instruction sequences on different data sets.

For example, SMPs, clusters and NUMA systems fit into this category. All distributed systems are also fall into this category.

Flynn's Taxonomy



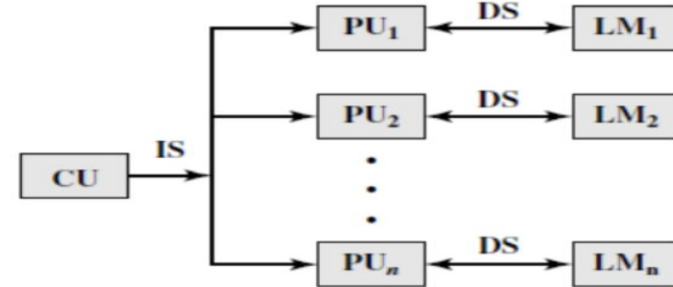
(a) SISD



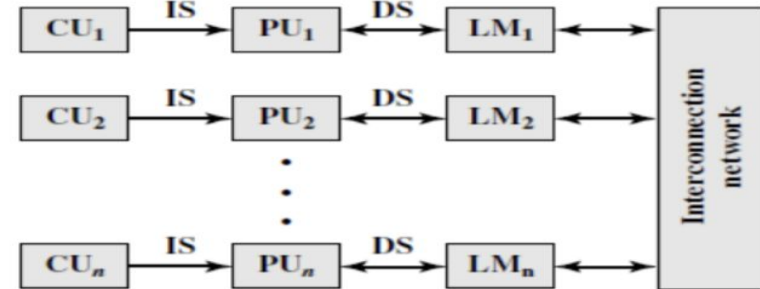
(c) MIMD (with shared memory)

CU = Control unit
 IS = Instruction stream
 PU = Processing unit
 DS = Data stream
 MU = Memory unit
 LM = Local memory

SISD = Single instruction,
 single data stream
 SIMD = Single instruction,
 multiple data stream
 MIMD = Multiple instruction,
 multiple data stream



(b) SIMD (with distributed memory)



(d) MIMD (with distributed memory)

Figure 17.2 Alternative Computer Organizations